

NLU Project — Opinion Mining with LoRA Fine-Tuning

Project Overview

This project compares a **base language model** against the **same model after fine-tuning** on the task of opinion mining from movie reviews.

Instead of simple sentiment classification (positive/negative), the model is asked to extract four features from each review:

- **Sentiment** — positive or negative
- **Emotions** — e.g. joy, anger, sadness, excitement, disappointment
- **Aspects** — what topics are mentioned: acting, plot, visuals, direction, music, humor
- **Intensity** — strong, moderate, or weak

The fine-tuning technique used is **LoRA (Low-Rank Adaptation)** — a method that keeps the original model frozen and only trains a small set of added parameters (~3.4% of the total), making it fast and memory-efficient enough to run on a free Kaggle GPU.

Dataset: IMDB movie reviews — 8,000 for training, 800 for testing

Model: Qwen2.5-0.5B (500M parameter language model, 4-bit quantized)

Hardware: Kaggle free tier — NVIDIA T4 GPU

Training time: ~45-55 minutes

Cell-by-Cell Overview

Cell 1 — Installation

Installs all required Python packages: `transformers`, `datasets`, `bitsandbytes`, `peft`, `trl`, `accelerate`, `scikit-learn`, `matplotlib`, and `seaborn`.

Cell 2 — Dependencies & Environment Setup

Imports all libraries and sets `CUDA_VISIBLE_DEVICES=0` to restrict the notebook to a single GPU. This prevents a known crash where PyTorch tries to split the quantized model across two GPUs, which bitsandbytes does not support.

Cell 3 — Dataset Preparation & Feature Labeling

Loads the IMDB dataset and selects 8,000 training and 800 test samples. Since IMDB only

provides sentiment labels, the other three features (emotions, aspects, intensity) are derived automatically from the review text using keyword matching and a scoring heuristic. Each sample is then formatted as an instruction-response pair that the model will learn from.

Cell 4 — Base Model Loading

Downloads and loads `Qwen2.5-0.5B` with 4-bit quantization to reduce memory usage. The model is pinned to a single GPU (`device_map={"": 0}`) and caching is disabled to prepare it for training.

Cell 5 — Evaluation Helpers

Defines three reusable functions used throughout the notebook:

- `parse_output()` — extracts the four fields from the model's raw text using regex
 - `evaluate_model()` — runs inference on a set of reviews and collects predictions
 - `score_results()` — computes the 5 evaluation metrics and prints a summary
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Cell 6 — Base Model Evaluation (Before Fine-Tuning)

Runs the unmodified base model on 150 test samples. This is the zero-shot baseline — the model has never been trained on this task. Results here represent what the model can do purely from its general pre-training.

Cell 7 — LoRA Configuration

Prepares the base model for training and attaches LoRA adapters to both the attention layers and the MLP feed-forward layers. Only these adapter weights (~17.6M parameters) will be updated during training — the rest of the model stays frozen.

Cell 8 — Training

Configures and runs the fine-tuning with `SFTTrainer`. Trains for 3 epochs over 8,000 samples using the paged AdamW optimizer and a cosine learning rate schedule. The trained adapter weights are saved at the end.

Cell 9 — Fine-Tuned Model Evaluation

Runs the same evaluation as Cell 6, on the same 150 test samples, with the same prompt — but now using the fine-tuned model. This produces the post-training scores to compare against the baseline.

Cell 10 — Side-by-Side Comparison & Plots

Prints full classification reports for both models and generates a 6-panel figure showing: a grouped bar chart of all 5 metrics, a gain chart showing improvement from LoRA, confusion matrices for sentiment on both models, and a summary box.

Cell 11 — Live Demo

Tests both models on 5 hand-picked reviews designed to expose real differences — negative reviews with positive-sounding words, multi-aspect reviews, very short reviews, and emotionally intense reviews with ALL-CAPS and exclamation marks. Outputs both predictions side by side and flags where the models disagree.

Results Summary

Metric	Base Model	After LoRA	Gain
Format compliance	97.3%	100.0%	+2.7%
Sentiment accuracy	70.7%	80.7%	+10.0%
Emotion hit rate	69.3%	69.3%	0%
Aspect hit rate	91.3%	100.0%	+8.7%
Intensity accuracy	32.0%	40.7%	+8.7%

The biggest gains are in sentiment (+10%) and aspect detection (+8.7%). Emotion hit rate did not improve because the base model already converges on reasonable emotions from its general training, and the dataset size was not large enough to shift it further. Intensity remains the hardest field due to the subjectivity of the labels.